

Week 11

Boost Converter PCB Update

Arduino shield connection, sensing, protection, and layout finalization

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This Week's Changes

- Switched Arduino connection from loose wires to a shield-style plug-in design
- Replaced the load resistor with battery connection pins
- Added input voltage and current sensors for real-time monitoring
- Added a zener diode to protect the feedback signal into the Arduino ADC
- Planned physical clearance for heatsinks on the MOSFET and diode
- Positioned all 4 Arduino header connectors using exact official coordinates
- Finalized PCB layout zones and measured board dimensions

Arduino: Shield-Style Connection

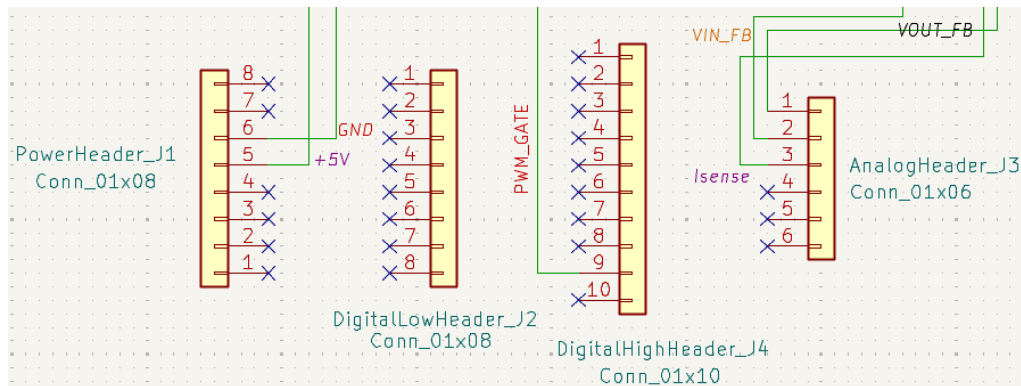
- Why: loose jumper wires near a 20kHz switching signal can pick up noise and interfere with each other
- Fix: PCB plugs directly onto the Arduino, like a shield — short, fixed connections, no loose wires

**Power
(8-pin)**

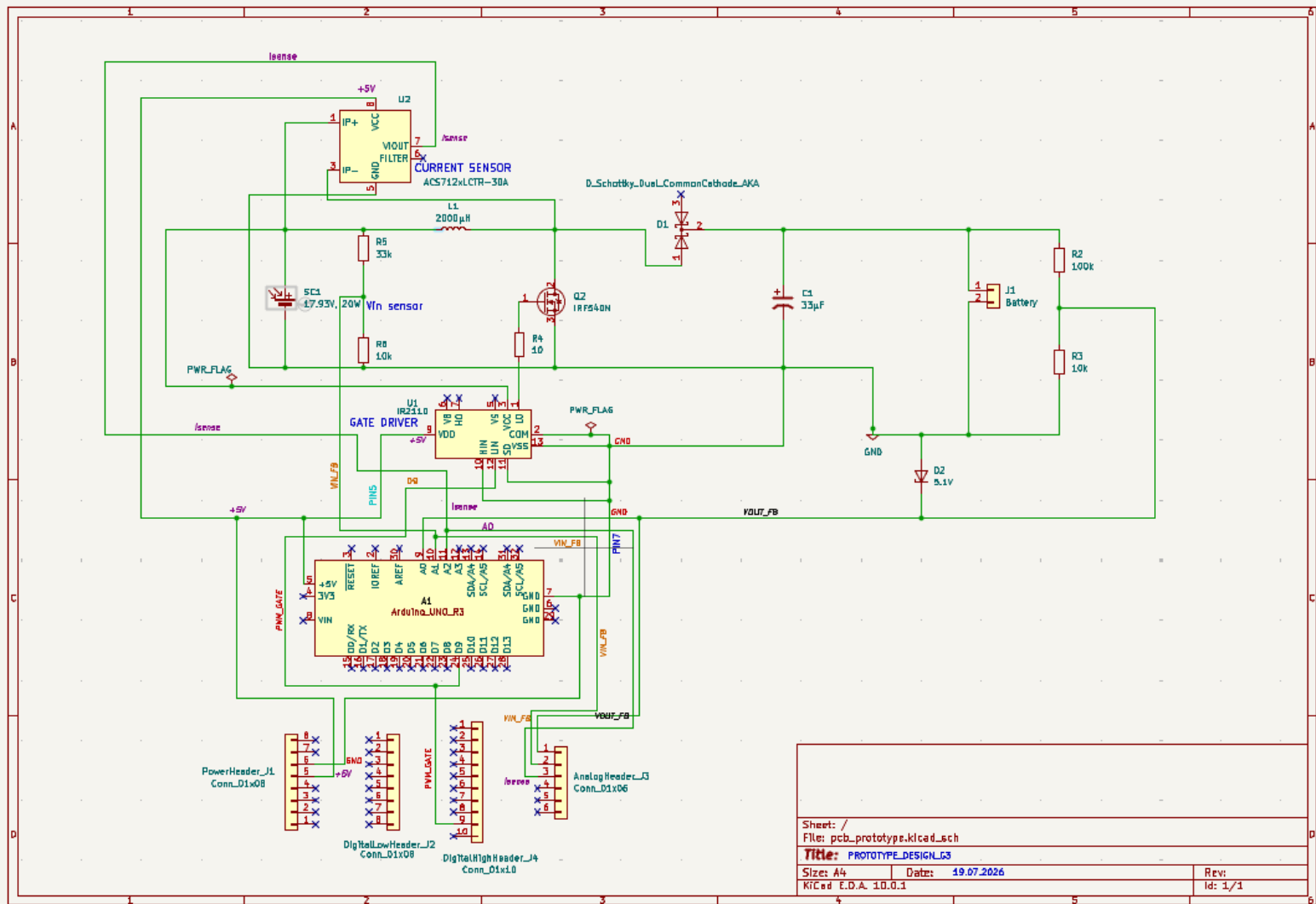
**Analog In
(6-pin)**

**Digital Low
(8-pin)**

**Digital High
(10-pin)**



4 header connectors placed at exact Arduino Uno spacing — grouped and locked as one unit



Voltage & Current Sensing + Protection

New sensors added

- Current sensor (ACS712) — inline between PV panel and inductor, feeds Arduino A2
- Vin voltage sensor — divider tapping the PV input, feeds A1
- Vout voltage sensor — existing R2/R3 divider, feeds A0

Protection added

- Zener diode (5.1V) placed between the Vout feedback line and GND
- Clamps the signal into A0, protecting the Arduino's ADC from voltage spikes

Heatsinks & Component Clearance

- Heatsinks needed on Q2 (IRF540N) and D1 (MBR10100CT) — both dissipate real power at switching frequency
- Not a schematic component — purely a PCB layout and mechanical planning consideration
- Kept 15-20mm clearance between Q2 and D1, so clip-on heatsinks don't collide
- Repositioned C1 (output capacitor) away from Q2 to avoid crowding the heatsink zone
- Insulating thermal pads planned if both heatsinks sit close together, in case TO-220 tabs are live
- WANT TO CONSIDER THE

Building the Shield-Style Connector

- Sourced the official Arduino Uno R3 Eagle CAD files (header pin coordinates)
- Placed 4 generic pin headers (Power, Analog In, Digital Low, Digital High) at the exact official spacing
- Grouped and locked the 4 headers, so relative spacing can't be accidentally disturbed
- Result: PCB plugs directly onto a real Arduino Uno, no loose wiring needed

Reference template used:

github.com/LNSD/Arduino-Shield-Template

PCB Layout Dimensions

Board size

Approx. 110mm x 100mm — sized to comfortably fit the power stage, gate drive, sensing, and header zones

Key clearances

Q2 to D1	15-20mm (heatsink clearance)
Gate driver (U1) to Q2 gate	under 20mm (short gate trace)
Power stage to feedback/sensing zone	25-30mm (noise isolation)
Component to board edge	2.5-4mm minimum
Header block to nearest component	20-25mm

PCB connections are in progress

